Conversion of Borneol to Camphor

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CHEM 2081-007

Introduction

A reflux apparatus was used to convert a racemic mixture of the secondary alcohol borneol to the ketone camphor via oxidation of the alcohol. Extraction methods were utilized to increase the optimum percent yield.

Materials and Methods

Approximately 0.30 g of racemic borneol and 0.30 mL of glacial acetic acid were combined in a 10 mL long-neck round-bottom flask. Then, a boiling chip and 1 mL of acetone were added to the flask to dissolve the mixture. A reflux apparatus was assembled and attached to the flask. A hot water bath was prepared and maintained at approximately 40 - 50°C, and the flask was placed in the water while the borneol dissolved. Thereafter, roughly 6 mL of bleach was added to the flask in a drop-wise manner over 35 minutes; the solution was gently swirled after each addition. Once all bleach had been added, the solution was allowed to sit in the warm water bath for 15 minutes to complete the reaction. The flask was removed from the water, cooled to room temperature, and then the solution was transferred to a centrifuge tube. The product was extracted by adding roughly 2 mL of dichloromethane (DCM) to the solution. Upon observation of the separation of layers, the bottom layer was rinsed over the top layer 3-4 times. A Pasteur pipette was used to extract the bottom layer and transfer it to a clean 10 mL beaker. The aqueous fractions were then transferred to another beaker, and the organic fractions (bottom layer) were put back into the centrifuge tube, and the extraction process was repeated. Once complete, approximately 2 mL of saturated sodium bicarbonate was added to the organic fractions to neutralize the DCM. The organic solution was then separated from the $NaHCO_3$ and transferred to a clean centrifuge tube. The organic solution was washed sequentially with 2 mL of 5% sodium bisulfite and 2 mL of water. The product was then isolated, transferred to a new

centrifuge tube, and dried with anhydrous sodium sulfate. Once dry, the solution was poured onto a pre-weighed watch glass and set over a warm water bath to evaporate off the DCM, leaving only the product. The watch glass was then weighed, and the yield was calculated. All organic fractions were disposed of appropriately, and all glassware was cleaned under the fume hood.

Results

Theoretical Yield of Camphor (g):	Experimental Yield of Camphor (g):	Percent Yield of Camphor (%):
0.300 g	0.275 g	91.7 %

Table 1: Theoretical and experimental yield of camphor and calculated percent yield of camphor.

Calculations

$$0.304\ g\ of\ borneol \times \frac{mol}{154.25\ g} = 1.97 \times 10^{-3}\ mols\ of\ borneol$$

$$1.97 \times 10^{-3}\ mols\ camphor \times \frac{152.23\ g}{mol} = 0.300\ g\ camphor^{**}$$

$$Percent\ Yield = \frac{0.275\ g}{0.300\ g}\ x\ 100 = 91.7\ \%$$

** = Theoretical Yield

Discussion

In this experiment, approximately 0.275 g of camphor was synthesized via the oxidation of 0.304 g of borneol. The borneol was dissolved in acetone and acetic acid. The addition of sodium hypochlorite solution allowed the reaction to proceed as it produced hypochlorous acid

that immediately reacted with the borneol in the solution. Once the reaction was complete and cooled to room temperature, dichloromethane was used to extract the product, and the solution was separated into two separate layers. The product is solvated in the DCM due to similar polarity properties between the two compounds. The bottom layer of each extraction was assumed to be the product/DCM layer due to DCM's higher density than the aqueous components. The theoretical yield of this experiment was 0.300 g of camphor, so the percent yield was calculated to be approximately 91.7%. One possible error in this experiment could be multiple extraction steps carried out using a Pasteur pipette, which is not very precise when collecting the product. Another possible error could be overheating the solution during reflux, resulting in an incomplete product. More than likely, the way the extractions were carried out was probably the most significant source of error. Using a separatory funnel in future attempts at this experiment would increase precision during extraction steps and decrease the loss of product.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University.

Exp. No. 6-3	Experiment/Subject Borneol to camphor	Date 11/16/23	
Name Kelli Furlow	Lab Partner	Locker/ Desk No.	Course & Section No. 2081 007

Borneol: ($C_{10}H_{18}O$)



MW = 154.25 g/mol

BP = 212°C

Density = 1.011 g/mL

Camphor: ($C_{10}H_{16}O$)



MW = 152.23 g/mol

BP = 209°C

Density = 0.99 g/mL

Acetic Acid: (CH_3COOH)



MW = 60.05 g/mol

BP = 118°C

Density = 1.05 g/mL

Acetone: (C_3H_6O)

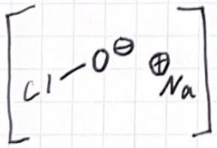


MW = 58.08 g/mol

BP = 56°C

Density = 0.791 g/mL

Sodium Hypochlorite: ($NaClO$)



MW = 74.44 g/mol

BP = 111°C

Density = 1.25 g/mL

Dichloromethane: (CH_2Cl_2)



MW = 84.93 g/mol

BP = 40°C

Density = 1.33 g/mL

Signature	Date	Witness/TA	Date

Note: Place fold-over back cover under copy sheet before writing

Exp. No. 6-3	Experiment/Subject Borneol to Camphor	Date 11/16/23	
Name Kelli Furlow	Lab Partner	Locker/ Desk No.	Course & Section No. 2081 007

Procedure:

#1) Combine 0.3g of borneol & 0.3 mL of acetic acid in a long-neck round-bottom 10 mL flask
- Add 1 mL of acetone to dissolve mixture & add a boiling chip

Dissolves in water
- Density of H_2O = 1.00 g/mL

0.304g borneol used

#2) Prepare a sand bath & begin heating a 100 mL beaker filled w/ ~75 mL of water
- Assemble a reflux apparatus & heat flask in the water at ~40-50°C
* If no observable dissolving of borneol is noted then add an additional 0.5 mL of acetone to the mixture

#3) Add ~6 mL of aqueous ~~NaHCO~~ NaClO sltn (bleach) over the top of the condenser in a dropwise manner
* Done over 35 minutes *

- After each addition swirl the mixture & allow for the rxn to proceed for 15 minutes once the last addition of bleach is done

½ mL/minute

#4) Allow the mixture to cool to room temperature & then add ~2 mL of methylene chloride to extract the camphor - perform this twice & place organic fractions in a centrifuge tube

- Bottom layer is the product
- Add DCM see two layers
- Organic sltn = only organic layer that is isolated

- Rinse bottom layer over top

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Exp. No. 6-3	Experiment/Subject Borneol to Camphor	Date 11/16/23	
Name Kelli Furlow	Lab Partner	Locker/ Desk No.	Course & Section No. 2031 007

#5) Then add ~2 mL of saturated $NaHCO_3$ to the organic layer - keep adding until presence of CO_2 bubbles is no longer noted - indicating neutralization of the DCM

#6) Wash the organic layer w/ 2 mL of 5% sodium bisulfite followed by 2 mL of water

#7) Dry also over anhydrous sodium sulfate

#8) Evaporate DCM over water bath w/ temp not exceeding 50°C
- Weigh collected crystals & determine % yield

- Drying agent transferred w/ product to watch glass

35.812g = watch glass

36.087g = watch glass + product

36.087g − 35.812g = 0.275g Camphor

$$\frac{0.304\text{g borneol}}{} \Big| \frac{\text{mol}}{154.25\text{g}} =$$

$$\frac{0.00197\text{ mol}}{} \Big| \frac{152.23\text{g}}{\text{mol}} =$$

Theoretical Yield of Camphor = 0.300g Camphor

Experimental Yield of Camphor = 0.275g

$$\% \text{ Yield} = \frac{0.275\text{g}}{0.300\text{g}} \times 100$$

% Yield = 91.7%

- Separatory Funnel may increase yield
- DCM is very volatile & may have evaporated prematurely
- Overheating the sample w/ too hot of a water bath

Signature	Date	Witness/TA	Date

Note: Place fold-over back cover under copy sheet before writing

Lab Performance

Exp 6 D3

Kelli Furlow | 007 | 402

Name: *Section:* *Room:* *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.	○ ○ ○ ○ ~~○~~ (5pts)
Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.	○ ○ ○ ○ ○ (5pts)
Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).	○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)
Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).	○(1pt) ○ (5pts) ○ (5pts)
Working independently or equal partner participation during the entire lab.	○ ○ ○ ~~○~~ ○ (5pts)
Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.	○ ○ ○ ○ ○ (5pts)
Maintain clean and organized personal space.	○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* 3